
Calibrating What Behavior Identifies: Conformal Reward Sets for Inverse Reinforcement Learning

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Abstract

1 Conformal inverse optimization (CIO) calibrates an uncertainty set around an
2 estimated objective, then optimizes robustly against it. We ask which parts of that
3 recipe survive the move to inverse reinforcement learning (IRL). The conformal
4 calibration transfers untouched; two other components break, both because behavior
5 identifies a reward only up to transformations that leave every policy’s value
6 unchanged. CIO scores a demonstration by an angle between reward coordinate
7 vectors, and a feature reparameterization that changes no reward value and no
8 demonstrator’s policy still moves the calibrated radius by 24° ; we replace the
9 angle by distance to a Bellman-feasible reward cone in the span seminorm of the
10 occupancy polytope, which is invariant by construction and still a linear program.
11 The calibrated event is also weaker than it looks: it certifies that *some* reward in
12 the set explains the new demonstrator (0.82 measured, at target 0.80), not that the
13 demonstrator’s own reward lies in it (0.44). Closing that gap takes identification
14 rather than more data—a certified bound η on inverse-fiber width—giving the
15 containment radius $R = \min\{2, 2q_\gamma + \eta\}$ and a sharp threshold $R \geq 1$ past which
16 the safety guarantee provably carries no decision content.

17 1 Introduction

18 IRL turns behavior into a reward estimate and then optimizes that estimate, discarding the uncertainty
19 left by the first step exactly where it matters: different rewards consistent with the same demonstrations
20 prescribe different policies. Conformal inverse optimization (CIO) is an appealing remedy for one-
21 shot inverse problems [Lin et al., 2024]. Observe a population of decision makers, fit a point estimate
22 on part of it, and on the rest measure how far that estimate sits from the set of objectives that would
23 rationalize each held-out decision; the conformal quantile of those distances defines an uncertainty set
24 with finite-sample coverage, and robust optimization over the set replaces the point prescription. The
25 recipe is modular and looks portable. We carried it into IRL and report what happened, component
26 by component (Table 1).

27 One component transfers for free, and it is not our contribution: once per-demonstrator scores are
28 defined, the rest of the calibration is split conformal prediction—rank the scores, take the $\lceil \gamma(N+1) \rceil$ -
29 th, invoke exchangeability [Vovk et al., 2005, Lin et al., 2024]. Sequential decision making changes
30 no step of that argument. The work is in the components on either side, and both break for the same
31 reason.

32 **Behavior does not fix reward coordinates.** Potential shaping, rewards on unreachable states, and
33 other additive directions change a reward vector while adding the same constant to every policy’s value,

Table 1: The CIO recipe, component by component, in IRL. Bold entries mark what had to change; the rest, including the conformal quantile itself, are inherited and used as such.

CIO ingredient	Counterpart here	Status
Point estimate; inverse-feasible set of a decision	any IRL fit; the Bellman cone $K(\pi)$, linear in θ after resolvent substitution	inherited
Score: angle to the estimate	span-seminorm distance to $K(\pi)$; invariant, still an LP	redesigned (§2, E1)
Split-conformal quantile	unchanged	inherited, textbook
Calibrated event $\Rightarrow$ parameter set	needs an inverse-fiber width η that behavior cannot supply	new assumption (§3, E2)
Robust decision	reference-relative robust MDP, a single LP; absolute version is unbounded	redesigned (§3)

so an angle between reward coordinate vectors spends its radius on distinctions no demonstration can reveal. This is not hypothetical: an invertible feature reparameterization leaving every scalar reward and every demonstrator’s policy pointwise unchanged still moves the calibrated angular radius by 24° (E1), so two mathematically identical problems get two different uncertainty sets. The fix is not a better estimator but an answer to the prior question—what does behavior distinguish?

Intersection is not containment. Split conformal certifies that *some* reward in the calibrated set makes the new demonstrator optimal; a robust prescription needs the demonstrator’s *own* reward to lie in the set. In one-shot inverse optimization the two nearly coincide, but in IRL a single policy is rationalized by a whole cone: at target 0.8 we measure the first event at 0.82 and the second at 0.44 (E2). Calibration cannot close that gap, because the missing quantity is identification, not statistics.

We therefore rebuild the score in decision-space geometry (§2) and add an explicit bridge from intersection to containment (§3), with a sharp boundary at which its own safety promise becomes vacuous. We do *not* claim that CP-IRL lowers true-reward regret against point-estimate IRL; the claim is narrower and, for anyone considering the same transplant, more useful: half of CIO’s pipeline moves to IRL unchanged, two components must be rebuilt, and one thing cannot be supplied by behavioral data at all.

Related work. CIO calibrates uncertainty for a latent objective [Lin et al., 2024] using split conformal prediction [Vovk et al., 2005]. Two lines are directly relevant: partial identifiability of reward sets in IRL [Skalse et al., 2023, Kim et al., 2024] and reward-comparison metrics, of which STARC [Skalse et al., 2024] subsumes EPIC [Gleave et al., 2021]. Our occupancy span is the canonical seminorm behind one STARC construction, so we claim no new reward metric: the contribution is calibrating Bellman cones *in* that geometry, plus the fiber-width bridge. Appendix B covers the rest.

2 An invariant score with exact intersection coverage

A tabular MDP $M = (S, A, P, \rho_0, \beta, \phi)$ is shared and known, with rewards $r_\theta(s, a) = \phi(s, a)^\top \theta$; demonstrator k draws $\hat{\theta}_k \sim P_\theta$, acts optimally under it, and we observe its deterministic policy π_k exactly. Let $M(P)$ be the polytope of normalized discounted occupancy measures, Φ the feature matrix, $V_\theta(\mu) := \theta^\top \Phi^\top \mu / (1 - \beta)$, and let $K(\pi) := \{\theta : Q_\theta^\pi(s, a) \leq Q_\pi^\pi(s, \pi(s)) \ \forall s, a\}$ be the unbounded Bellman cone of rewards making π optimal; fixing π makes it exactly $|S|(|A| - 1)$ halfspaces (Proposition 3).

Assumption 1 (Exchangeable calibration unit). *The pairs $(\hat{\theta}_k, \pi_k)$, $k = 1, \dots, N, N+1$, are i.i.d. (hence exchangeable), and M (in particular P) is fixed across all demonstrators and the deployment instance.*

One demonstrator is thus one calibration point, playing the role of CIO’s (decision, context) pair.

The occupancy-difference span. For $v \in \mathbb{R}^d$ let

$$\|v\|_{\mathcal{D}} := \max_{\mu \in M(P)} V_v(\mu) - \min_{\mu \in M(P)} V_v(\mu) = V_v^*(\rho_0) + V_{-v}^*(\rho_0),$$

two value-iteration solves. Read $\|v\|_{\mathcal{D}}$ as: how much can the reward direction v change any policy's value? Directions no policy can exploit score zero.

Proposition 1 (Properties of $\|\cdot\|_{\mathcal{D}}$). $\|\cdot\|_{\mathcal{D}}$ is a seminorm with $\|v\|_{\mathcal{D}} = 0$ iff $v \in \mathcal{A}(P) := \{a : a^\top \Phi^\top (\mu - \nu) = 0 \forall \mu, \nu \in M(P)\}$, the additively null subspace, which contains the potential-shaping subspace S_{pot} and can be strictly larger. For any θ, θ' with $\mu_{\theta'}$ optimal for θ' , $\text{Reg}_\theta(\pi_{\theta'}) := V_\theta(\mu_\theta^*) - V_\theta(\mu_{\theta'}) \leq \|\theta - \theta'\|_{\mathcal{D}}$. Finally $\|\cdot\|_{\mathcal{D}}$ is exactly invariant under $\phi \mapsto A\phi$, $\theta \mapsto A^{-\top}\theta$ for invertible A .

The regret constant is 1—no Cauchy–Schwarz slack, no feature-norm Lipschitz factor—because $\|\cdot\|_{\mathcal{D}}$ is defined as the very value spread it must bound. Behavior also cannot identify positive reward scale, which we remove by normalizing rather than by building quotient coordinates: writing $\theta^\circ := \theta/\|\theta\|_{\mathcal{D}}$, the map $d_o(\theta, \theta') := \|\theta^\circ - \theta'^\circ\|_{\mathcal{D}}$ is a pseudometric on $[0, 2]$ that vanishes exactly when $\theta' = c\theta + a$ with $c > 0$, $a \in \mathcal{A}(P)$, and still bounds normalized regret (Proposition 4). So d_o compares rays of policy-value differences.

Score and calibrated set. With $\bar{\theta}$ an IRL point estimate fit on a disjoint training split, define

$$s_k := \min_{\theta \in K(\pi_k)} \|\theta - \bar{\theta}^\circ\|_{\mathcal{D}}, \quad B_q := \{\theta : \|\theta - \bar{\theta}^\circ\|_{\mathcal{D}} \leq q\}.$$

Since $\|\cdot\|_{\mathcal{D}}$ has a linear-program epigraph, s_k is a single LP in $d + 2|S|$ variables, with no second-order cone anywhere; the Euclidean ball CIO uses instead would make it an SOCP and destroy reparameterization invariance. Because $0 \in K(\pi_k)$ we have $s_k \in [0, 1]$, and both score and quantile are unchanged under $\bar{\theta} \mapsto c\bar{\theta} + a$ for $c > 0$, $a \in \mathcal{A}(P)$.

Theorem 1 (Decision-aware coverage, exact equivalence). *Calibrate q_γ as the τ -th smallest s_k among N_{val} demonstrators, $\tau = \lceil \gamma(N_{\text{val}} + 1) \rceil$ (sentinel $+\infty$ if $\tau = N_{\text{val}} + 1$). For a fresh i.i.d. demonstrator, $s_{\text{new}} \leq q_\gamma \iff K(\pi_{\text{new}}) \cap B_{q_\gamma} \neq \emptyset$, and $\Pr[K(\pi_{\text{new}}) \cap B_{q_\gamma} \neq \emptyset] \geq \gamma$.*

The probability statement is the standard split-conformal rank argument [Vovk et al., 2005] applied verbatim. What is specific here is the equivalence to its left, immediate from s_k being a minimum: coverage of the score is intersection coverage of the set, with no lossy conversion in between.

3 From intersection to containment, and the price of the bridge

The Bellman cone is not an equivalence class: even after removing scale and additive symmetries it can contain rewards with genuinely different policy orderings. Let $F^\circ(\pi) := \{\theta \in K(\pi) : \|\theta\|_{\mathcal{D}} = 1\}$ be the normalized inverse fiber, modulo the zero directions of $\|\cdot\|_{\mathcal{D}}$.

Assumption 2 (Certified inverse-fiber width). *There is a known $\eta \in [0, 2]$ with $\sup_{u, v \in F^\circ(\pi)} \|u - v\|_{\mathcal{D}} \leq \eta$ almost surely for the deployment population, and each latent reward is behaviorally nonnull.*

This is the extra information needed to turn intersection into containment. It must come from side information, multiple environments, or active comparisons; with none of those, the only universal certificate is $\eta = 2$.

Theorem 2 (Two-stage containment and safe prescription). *Set $R := \min\{2, 2q_\gamma + \eta\}$ if $q_\gamma < 1$ and $R := 2$ otherwise. Then $\Pr[\|\hat{\theta}_{\text{new}}^\circ - \bar{\theta}^\circ\|_{\mathcal{D}} \leq R] \geq \gamma$, and if μ_R maximizes worst-case value over B_R relative to a fixed reference $\mu_{\text{ref}} \in M(P)$, then $\Pr[V_{\hat{\theta}_{\text{new}}}(\mu_R) \geq V_{\hat{\theta}_{\text{new}}}(\mu_{\text{ref}})] \geq \gamma$.*

The factor 2 is the price of keeping Stage I convex. The reference is not cosmetic: B_R is unbounded—it contains all of $\bar{\theta}^\circ + \mathcal{A}(P)$ —so worst-case absolute value is $-\infty$, and only a comparison between two occupancy measures is insensitive to directions with no behavioral content. Relative to μ_{ref} the inner minimum is the center advantage minus $R\gamma_{D_F}(x)$, where γ_{D_F} is the Minkowski gauge polar to $\|\cdot\|_{\mathcal{D}}$; it has an LP epigraph, so the deployed problem is a single linear program with penalty normalized to $[0, 1]$ (Appendix A).

Proposition 2 (Sharp degeneracy threshold). *For the unit-span center, if $R \geq 1$ the optimal relative robust value is zero, attained by μ_{ref} ; if $R > 1$ every optimizer is feature-equivalent to the reference. A nontrivial safe improvement therefore requires $2q_\gamma + \eta < 1$.*

This turns vacuity into an auditable output: the implementation reports q_γ , η , R , and whether $R < 1$. All experiments center μ_{ref} at the calibration population's mean occupancy, which asks the question CP-IRL is about—does this policy beat what a random draw from the population achieves?

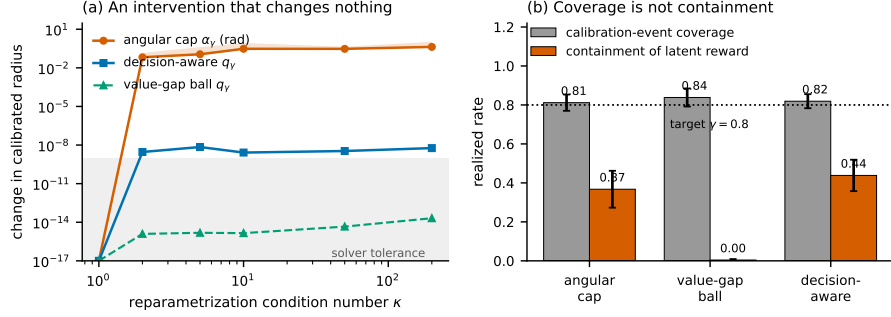


Figure 1: The two failures. **(a)** A random invertible feature reparameterization of condition number κ leaves every pointwise reward and policy unchanged, yet the inherited angular cap moves; the unit-span score stays fixed to solver tolerance (shading: mean to worst over 10 seeds). **(b)** At $\gamma = 0.8$ every calibration event attains marginal coverage, but containment of the latent reward ray at the same radius does not. Error bars are 95% t intervals over 30 environment–seed blocks.

4 Experiments

Both experiments are mechanism tests, not a claim that robustification lowers true-reward regret. We use random tabular MDPs, a Voronoi-feature gridworld, and Objectworld ($|S| \leq 36$, $d \leq 5$), fitting MaxEnt and Bayesian IRL on 20 training demonstrators, calibrating on 20, and testing on 20 more, over 10 environment seeds (1,200 test rewards). Intervals are 95% Student- t over the 30 independent environment–seed blocks; details in Appendix C.

E1: a reward-preserving intervention moves the angular radius. We draw one random invertible A at each $\kappa \in \{1, 2, 5, 10, 50, 200\}$ and transform $\phi \mapsto A\phi$, $\theta \mapsto A^{-\top}\theta$. Coordinates change; every scalar reward is preserved to 2.2×10^{-14} , so a behaviorally meaningful radius should not move. At $\kappa = 200$ the angular radius changes by 0.424 radians (24.3°) on average and 1.071 radians in the worst seed, while the normalized span radius changes by at most 2.0×10^{-8} (Figure 1a). This is a falsification test, not a favorable data distribution: the two problems are mathematically identical.

E2: intersection and containment come apart. At $\gamma = 0.8$, pooled calibration-event rates are 0.812 ± 0.042 (angular), 0.838 ± 0.046 (value-gap), and 0.819 ± 0.036 (decision-aware); the first and third are exactly inverse-set intersection. Latent-ray containment at the same radius is only 0.368 ± 0.095 and 0.438 ± 0.081 respectively, and the value-gap-to-Euclidean conversion contains 0.004 ± 0.005 (Figure 1b). Conformal ranking behaves exactly as the theory says; the stronger event that robust safety needs does not follow from q_γ alone. The calibrated radius has pooled mean 0.316, so by Proposition 2 a certificate would need $\eta < 0.37$ on average to leave anything to decide—and the universal $\eta = 2$ leaves nothing (Appendix C).

5 What this means

Metric choice is real and measurable (E1), but a conformal intersection radius is not yet a CIO uncertainty set for the latent reward (E2). Assumption 2 names the missing ingredient and Proposition 2 makes its required strength checkable at run time; the universal diameter 2 proves containment but returns the reference, which is honest safety with no decision content rather than an algorithmic failure. The open problem is thus not another IRL point estimator but a mechanism certifying a small η , and our exchangeability unit—one demonstrator, one exactly observed optimal policy, a shared known tabular MDP—is itself demanding.

The pattern generalizes: whenever an inverse problem is partially identified, a calibrate-then-prescribe pipeline must separately answer in which geometry the score is measured, how the quantile is computed, and what the calibrated event implies about the latent parameter. Conformal prediction answers only the second. CIO can leave the others implicit because one-shot inverse-feasible sets are thin; IRL is the regime where they are not.

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A Deferred results: robust geometry and the inherited angular construction

This appendix states the robust-program geometry referred to in §2 and collects the CIO-inherited pipeline that §2 replaces, since the experiments compare against it.

Theorem 3 (Concavity under reward-only ambiguity; failure with dynamics ambiguity). *(a) For any nonempty compact $C \subseteq \mathbb{R}^d$, if P is fixed and known then $\max_{\mu \in M(P)} \min_{\theta \in C} \theta^\top \Phi^\top \mu$ is a concave maximization over the fixed polytope $M(P)$, hence a convex outer problem whenever C admits a tractable linear-minimization oracle. (b) The guarantee need not survive when ambiguity couples reward and transition: there is a 2-state, 2-action MDP and a convex one-dimensional non-product ambiguity set over (θ, P) whose worst-case value function is not concave.*

Part (b) is existential (Appendix D.2): rectangular and other structured cases remain tractable, and failure of concavity is not a hardness proof.

Proposition 3 (Resolvent-based score reduction). *Fix π ; let $P_\pi(s, s') = P(s, \pi(s), s')$, let Φ_π be the feature matrix restricted to π 's actions, and let $W_\pi := (I - \beta P_\pi)^{-1} \Phi_\pi$. Then $V_\theta^\pi = W_\pi \theta$ and $Q_\theta^\pi(s, a) = \Psi_\pi(s, a)^\top \theta$ with $\Psi_\pi(s, a) := \phi(s, a) + \beta P(s, a, \cdot) W_\pi$, both linear in θ , so*

$$K(\pi) = \{\theta : \Psi_\pi(s, a)^\top \theta \leq \Psi_\pi(s, \pi(s))^\top \theta \forall s, a\}$$

exactly. The angular baseline additionally uses $\hat{\Theta}_2(\pi) = K(\pi) \cap \{\theta : \|\theta\|_2 \leq 1\}$. Both have d decision variables instead of $d + |S|$ and $|S|(|A| - 1)$ nontrivial linear inequalities, after one resolvent solve per distinct policy.

Proposition 4 (Projective span geometry). *$d_\circ(\theta, \theta') := \|\theta^\circ - \theta'^\circ\|_{\mathcal{D}}$, with $\theta^\circ := \theta / \|\theta\|_{\mathcal{D}}$, is a pseudometric with range $[0, 2]$, invariant to independent positive rescaling, to addition of elements of $\mathcal{A}(P)$, and to invertible feature reparameterization. For nonnull rewards $d_\circ(\theta, \theta') = 0$ iff $\theta' = c\theta + a$ with $c > 0$, $a \in \mathcal{A}(P)$; and $\text{Reg}_\theta(\pi_{\theta'}) / \|\theta\|_{\mathcal{D}} \leq d_\circ(\theta, \theta')$.*

Angular calibration. For this inherited baseline only, we replace every nonzero point estimate by its Euclidean normalization and hence use $\|\bar{\theta}\|_2 = 1$; the decision-aware construction instead uses intrinsic unit-span normalization. With the score $c_k = \max_{\theta \in \hat{\Theta}_2(\pi_k)} \theta^\top \bar{\theta}$ of Proposition 3:

Theorem 4 (Coverage). *Under Assumption 1, let $\tau = \lceil (N_{\text{val}} + 1)\gamma \rceil$. If $\tau \leq N_{\text{val}}$, let q_γ be the τ -th largest score among $c_1, \dots, c_{N_{\text{val}}}$; if $\tau = N_{\text{val}} + 1$, set the sentinel $q_\gamma = -1$. With $\alpha_\gamma := \arccos(q_\gamma)$,*

$$\mathbb{P}(\hat{\Theta}_{2, \text{new}} \cap C(\bar{\theta}, \alpha_\gamma) \neq \emptyset) \geq \gamma.$$

This is an intersection-nonemptiness guarantee, not a containment guarantee: it does not assert $\theta_{\text{new}}^ \in C(\bar{\theta}, \alpha_\gamma)$.*

The proof (Appendix D.8) transplants CIO's exchangeability argument onto these scores. A marginal-coverage simulation against the actual calibration code (150 trials per (γ, N_{val}) cell) confirms coverage $\geq \gamma$ at every tested combination; tied scores from finite populations are the likely source of the mild over-coverage beyond the rank argument's $1/(N_{\text{val}} + 1)$ slack. The sentinel returns the whole unit ball and is valid but uninformative; more generally, because the zero reward is Bellman-feasible for every policy, any cap with $\alpha \geq \pi/2$ makes the intersection event automatic, so coverage alone never establishes reward identification.

Theorem 5 (Regret bound under a reward-identification bridge). *Let $\nu(\mu) := \|\Phi^\top \mu\|_2 / (1 - \beta) \leq \bar{\nu} := \|\phi\|_{2, \infty} / (1 - \beta)$. Let θ_{new}^* be a unit-norm deployment reward, μ_{CIRL} maximize the worst-case value over $C = C(\bar{\theta}, \alpha)$, and μ^* be optimal for θ_{new}^* . Suppose (i) $\alpha \in [0, \pi/2]$ and (ii) compatible-reward accuracy: some $\theta_1 \in \hat{\Theta}_{2, \text{new}} \cap C$ has $\|\theta_1 - \theta_{\text{new}}^*\|_2 \leq \eta$. Then*

$$\text{AOG} := V_{\theta_{\text{new}}^*}(\mu^*) - V_{\theta_{\text{new}}^*}(\mu_{\text{CIRL}}) \leq (\eta + 2 \sin \alpha)(\nu(\mu^*) + \nu(\mu_{\text{CIRL}})) \leq 2(\eta + 2 \sin \alpha) \bar{\nu}.$$

Theorem 2 supplies the analogous bridge in the intrinsic geometry when the normalized inverse-fiber diameter is certified.

Proposition 5 (The calibrated robust MDP is a regularized MDP). *Let $\|\bar{\theta}\|_2 = 1$, $\alpha \in [0, \pi/2]$. Write $x = a(x)\bar{\theta} + x_\perp$ with $a(x) = \bar{\theta}^\top x$ and $b(x) = \|x_\perp\|_2$. Then*

$$\min_{\theta \in C(\bar{\theta}, \alpha)} \theta^\top x = \begin{cases} -\|x\|_2, & \text{if } a(x) \leq -\|x\|_2 \cos \alpha, \\ a(x) \cos \alpha - b(x) \sin \alpha, & \text{otherwise,} \end{cases}$$

so whenever every $\mu \in M(P)$ earns nonnegative estimated reward the robust policy solves $\max_{\mu \in M(P)} \{\bar{\theta}^\top \Phi^\top \mu - \tan \alpha \cdot b(\Phi^\top \mu)\}$.

Under the sign condition, then, CP-IRL is the point-estimate MDP with an ℓ_2 penalty at weight $\tan \alpha_\gamma$, fixed by the conformal quantile rather than tuned: $\alpha_\gamma \rightarrow 0$ recovers the point estimate and $\alpha_\gamma \rightarrow \pi/2$ refuses any off-direction feature component. Outside the sign condition the piecewise formula, not the regularizer reading, is the objective. The same form prices the hedge before deployment: writing μ_{plain} for the point-estimate optimum,

$$0 \leq \bar{\theta}^\top \Phi^\top \mu_{\text{plain}} - \bar{\theta}^\top \Phi^\top \mu_{\text{CIRL}} \leq \tan \alpha_\gamma \cdot b(\Phi^\top \mu_{\text{plain}}),$$

computable without ground truth. It bounds the loss in *estimated* reward only.

The value-gap repair and why it is incomplete. Replacing the angular score by the suboptimality gap under θ itself, $c_k := V_{\bar{\theta}}(\mu_{\bar{\theta}}^*) - V_{\bar{\theta}}(\mu_{\pi_k})$, needs one value-iteration solve plus one policy evaluation per demonstrator and no optimization over θ .

Lemma 1 (Shaping directions are exactly redundant on $M(P)$). *For any shaping direction c (via potential Φ_{pot}) and any $\mu \in M(P)$, $c^\top \Phi^\top \mu = -(1 - \beta)\rho_0^\top \Phi_{\text{pot}}$, a constant independent of μ .*

Theorem 6 (Shaping invariance of the value-gap score). *Let $\bar{\theta}' = \bar{\theta} + c$ for $c \in S_{\text{pot}}$. Then $\bar{\theta}'$ induces the same optimal occupancy measure as $\bar{\theta}$, automatically by Lemma 1 and with no extra hypothesis, and $c_k(\bar{\theta}') = c_k(\bar{\theta})$ for every π_k .*

Converting this score to a ball $C_{\text{ball}}(\bar{\theta}, \rho)$ with $\rho := q_\gamma/\bar{\nu}$ uses the Lipschitz constant of Theorem 5, which is where the construction fails: the bound licenses gap $\leq \text{const} \times \text{distance}$ and never the reverse, so q_γ alone certifies nothing about the set (measured containment 0.003, Figure 1b). Its robust program is at least unconditional:

Proposition 6 (The value-ball cap needs no case split). *For any $\rho \geq 0$ and x , $\min_{\|\theta - \bar{\theta}\|_2 \leq \rho} \theta^\top x = \bar{\theta}^\top x - \rho\|x\|_2$, so $\max_{\mu \in M(P)} [\bar{\theta}^\top \Phi^\top \mu - \rho\|\Phi^\top \mu\|_2]$ is the point-estimate MDP with a fixed-weight ℓ_2 penalty, for every $\bar{\theta}$ and ρ , with no sign condition.*

A tie-breaking pitfall in the degenerate separation example. A direct transplant of CIO’s separation example does not survive inspection. In a one-state, two-action MDP with $\phi_0 = (0, -1)$, $\phi_1 = (-u, 0)$ and demonstrator rewards on the first-quadrant arc, the exact-fit LP estimator returns the tie direction $\theta_u = (1, u)/\sqrt{1 + u^2}$, under which both actions are optimal. Every score is then $c_k = 1$, calibration returns $\alpha_\gamma = 0$, and the cap collapses to $\{\theta_u\}$, under which every action mixture has equal value. The apparent separation is produced by the solver’s tie-breaking rule, not by robustification; a valid separation needs a strictly positive radius and a unique robust optimizer. The same degeneracy explains why projective experiments must audit whether a fitted center has positive occupancy span.

B Extended related work

CIO calibrates uncertainty for a latent objective [Lin et al., 2024] using split conformal prediction [Vovk et al., 2005]; a parallel IRL line studies partial identifiability and feasible reward sets [Skalse et al., 2023, Kim et al., 2024, Poiani et al., 2024, Lazzati and Metelli, 2025, Kim et al., 2026] under observation models different from ours. Reward-comparison metrics ask which distance between rewards is meaningful: EPIC [Gleave et al., 2021] cancels shaping, DARD [Wulfe et al., 2022] restricts to realizable transitions, and STARC [Skalse et al., 2024] unifies both. Our occupancy span is the canonical seminorm behind one STARC construction, so the contribution is not a new reward metric: it is calibrating Bellman cones *in* that geometry, plus the explicit fiber-width bridge. Robust MDPs classically recover tractability through rectangularity [Iyengar, 2005, Wiesemann et al., 2013]; Gadot et al. [2024] solve the fixed-kernel L_p -ball reward case. We are not aware of prior work conformally calibrating an inverse-feasible reward set from a demonstrator population.

C Additional experiments

Setup detail. E1–E2 use random tabular MDPs, a Voronoi-feature gridworld, and Objectworld, each with $|S| \leq 36$ and $d \leq 5$. We fit MaxEnt IRL and Bayesian IRL by Metropolis–Hastings on 20 training demonstrators, calibrate on 20, and evaluate on 20 fresh demonstrators, for 10 environment seeds: 60 fitted/calibrated runs and 1,200 test rewards. Draws within a run share a center and radius,

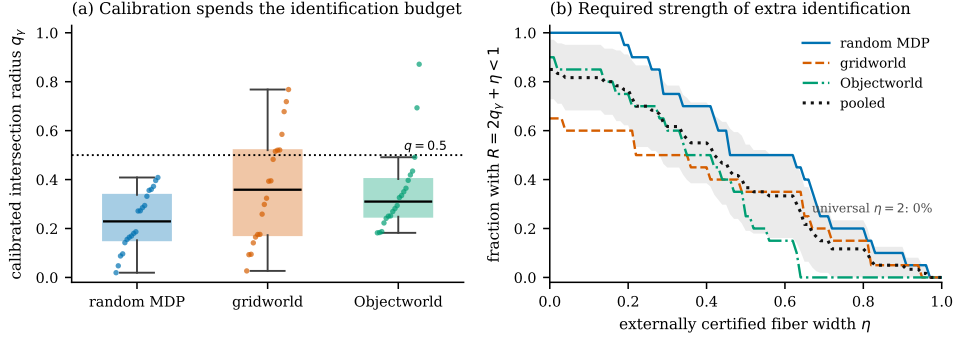


Figure 2: Identification-budget frontier for the numbers quoted in §4. **(a)** Distribution of the dimensionless Stage-I radius over 20 runs per environment. Even a zero-width fiber can be nontrivial only below $q_\gamma = 0.5$. **(b)** For each hypothetical certified width η , the fraction of runs satisfying $2q_\gamma + \eta < 1$. The curve does not estimate or validate η ; it states how strong an independent identification mechanism must be. Shading is a 95% t interval over 30 environment–seed blocks.

so intervals are 95% Student- t over the 30 independent environment–seed blocks after averaging the two estimators within each block. Computation used a 20-core laptop, no GPU, and no commercial solver.

The identification budget (E3). The calibrated q_γ has pooled mean 0.316, median 0.289, and range $[0.019, 0.871]$, so by Proposition 2 a certified width must satisfy $\eta < \max\{1 - 2q_\gamma, 0\}$ for the safe prescription to differ from the reference. With hypothetical certificates $\eta = 0, 0.2$, and 0.5 , respectively 85.0%, 76.7%, and 36.7% of runs remain nontrivial, and the environment matters: at $\eta = 0.2$ the fractions are 95% for random MDPs, 60% for gridworld, and 75% for Objectworld. The universal certificate $\eta = 2$ instead gives $R = 2$ in every run, expanded containment 1.000, reference dominance 1.000, and nontriviality 0.000—a theorem/implementation check, not a useful policy, since the deployed occupancy is exactly the reference (Figure 2).

Projective-center eligibility. We refit the exact-fit suboptimality-loss LP on the same 30 environment–seed training populations as the main audit. Its center has positive occupancy span in 29 cases and is additively null in one gridworld case. The latter has no unit-span representative. The MaxEnt and Bayesian estimators used in E2–E3 are nonnull in every reported run. This check is a domain-of-definition audit, not a comparison of estimator accuracy.

Legacy policy-regret comparison. Before introducing the certified fiber-width bridge, the exploratory three-estimator audit deployed each method at its Stage-I radius. Angular robustification increased true-reward regret in all nine environment–estimator cells, by factors 1.12–19.49. The value-gap ball was $0.72\text{--}1.86\times$ the point-estimate regret and improved three cells; the decision-aware radius improved five. These policies are not the Stage-II policy of Theorem 2, so the table diagnoses geometry and objective mismatch rather than demonstrating safe improvement.

Concentration baseline. Exact conformal calibration is tighter than a DKW bound everywhere tested; on the unbounded value-gap score DKW inflates true-reward regret $4\text{--}460\times$ over the point estimate, because the same additive slack $\sqrt{\log(2/\delta)/(2N)}$ shifts a raw-value threshold far more than a cosine in $[-1, 1]$.

Scale. Rerunning the full gridworld pipeline at $|S| = 225$, $d = 7$ ($6.25\times$ more states, two more reward dimensions) leaves the angular baseline’s negative result unchanged: its robust regret exceeds the plain estimate’s for all three estimators (LP-IRL $1.03 \rightarrow 3.19$, MaxEnt $1.01 \rightarrow 2.88$, Bayesian $0.06 \rightarrow 0.36$).

RLHF relabeling. On a “response selection” MDP whose population is annotators rather than task demonstrators, with the calibration and robust-MDP code unchanged, coverage meets or exceeds

target across $\gamma \in \{0.5, 0.7, 0.8, 0.9\}$ (0.947–0.994, 8 seeds, one configuration). Across four annotator-population configurations (4–10 responses, 3–6 attributes, 15–60 annotators) and 30 seeds at $\gamma \in \{0.7, 0.8, 0.9\}$, mean coverage per cell is 0.966–0.991 with at least 93% of seeds per cell meeting target, consistent with the marginal (not per-seed) nature of the guarantee. The gap above target likely comes from the same score-atom over-coverage as Theorem 4, sharpened by small populations.

Beyond linear rewards. $\|\cdot\|_{\mathcal{D}}$ and s_k are instances of an integral probability metric $\sup_{f \in \mathcal{F}_\psi} [\mathbb{E}_{\mu_\theta^*}(f) - \mathbb{E}_{\mu_k}(f)]$, with $\mathcal{F}_\psi = \{\theta\}$ giving the value-gap score, $\mathcal{F}_\psi = \{\text{linear}, \|\theta\|_2 \leq 1\}$ a closed-form dual-norm score, and richer classes (an RKHS ball, a bounded neural discriminator as in GAIL/AIRL) requiring adversarial training. On exact (non-sampled) occupancy measures, projected gradient ascent recovers the linear closed form to solver precision, and a strictly richer critic class, which contains the linear one and therefore weakly dominates it, finds 3–4 $\times$ more separation in every tested case. Whether a finite-trajectory *estimate* concentrates tightly enough, as $\mathcal{F}_\psi$ grows, for calibration to remain non-vacuous is the open piece with no directly reusable existing theory.

Reproduction. From the repository root, the main artifacts are generated by `experiments/run_reparam_invariance.py`, `experiments/run_containment_audit.py`, and `experiments/run_center_validity.py`; the two figures are generated by `paper/make_fig1_metric.py` and `paper/make_fig2_identification_frontier.py`. All scripts set NumPy random generators from the recorded integer seed. The audit JSON stores the schema version, split sizes, target γ , environment, estimator, and seed in every row.

D Proofs

Every identity and inequality stated in the main text is also checked numerically, independently of the library routines it describes, by `paper/verify_formulas.py`: the two-solve form of $\|\cdot\|_{\mathcal{D}}$ against brute-force enumeration of $M(P)$ ’s vertices, the seminorm axioms, the regret bound, reparametrization invariance, the zero set $\mathcal{A}(P)$, Lemma 1, the piecewise cap minimum of Proposition 5, the cap diameter and Lipschitz constant of Theorem 5, and the gauge form of the deployed robust objective.

D.1 Proof of Proposition 3

The value function V_θ^π solves the linear Bellman equation:

$$V_\theta^\pi = \Phi_\pi \theta + \beta P_\pi V_\theta^\pi.$$

Since $\beta \in [0, 1]$ and P_π is a stochastic matrix, its spectral radius satisfies $\rho(\beta P_\pi) \leq \beta < 1$, so $I - \beta P_\pi$ is invertible via a Neumann series, yielding:

$$V_\theta^\pi = (I - \beta P_\pi)^{-1} \Phi_\pi \theta = W_\pi \theta.$$

Substituting V_θ^π into the Bellman equation for the action-value function $Q_\theta^\pi(s, a)$ gives:

$$\begin{aligned} Q_\theta^\pi(s, a) &= \phi(s, a)^\top \theta + \beta P(s, a, \cdot) V_\theta^\pi \\ &= \phi(s, a)^\top \theta + \beta P(s, a, \cdot) W_\pi \theta \\ &= \Psi_\pi(s, a)^\top \theta, \end{aligned}$$

where $\Psi_\pi(s, a) := \phi(s, a) + \beta P(s, a, \cdot) W_\pi$, which is linear in θ as claimed.

By the policy improvement theorem, π is Bellman-optimal for reward θ if and only if

$$Q_\theta^\pi(s, a) \leq Q_\theta^\pi(s, \pi(s)) \quad \forall s \in S, \forall a \in A.$$

Substituting the linear expression for $Q_\theta^\pi(s, a)$ yields exactly the stated cone:

$$K(\pi) = \{\theta : \Psi_\pi(s, a)^\top \theta \leq \Psi_\pi(s, \pi(s))^\top \theta \quad \forall s, a\},$$

with no relaxation since the substitution is an equality.

The calibration score

$$c_k = \max_{\theta \in \bar{\Theta}_2(\pi_k)} \theta^\top \bar{\theta}$$

is then a linear objective over a set of $|S||A|$ linear inequalities plus the unit-ball constraint in $\theta \in \mathbb{R}^d$ alone: an SOCP of size d , independent of $|S|$ once Ψ_{π_k} is precomputed. ■

353 D.2 Proof of Theorem 3

354 (a) Define $g_C(x) := \min_{\theta \in C} \theta^\top x$. This is a pointwise infimum of the affine family $\{x \mapsto \theta^\top x : \theta \in C\}$. Since a pointwise infimum of affine functions is concave by definition for any nonempty C (convex, nonconvex, or discrete; concavity of g_C requires no assumption on C 's shape), composing 356 g_C with the linear map $\mu \mapsto \Phi^\top \mu$ preserves concavity:

$$\mu \mapsto g_C(\Phi^\top \mu) = \min_{\theta \in C} \theta^\top \Phi^\top \mu$$

358 is concave in μ .

359 Since $M(P)$ is a fixed polytope that does not depend on θ (the transition kernel P is known and 360 fixed), the ambiguity set never touches the feasible region, only the objective. Therefore,

$$\max_{\mu \in M(P)} g_C(\Phi^\top \mu)$$

361 is a concave maximization over a fixed compact convex set for *any* nonempty compact C . This 362 establishes the convex geometry; efficient solution also requires an efficient representation or linear- 363 minimization oracle for C . No Bellman recursion or rectangularity assumption enters the geometric 364 argument.

365 (b) Let state s_0 be active and s_1 absorbing with zero reward. At s_0 , action a_0 earns zero and moves to 366 s_1 ; action a_1 earns $1 + t$ and returns to s_0 with probability $1 - t$ (otherwise moving to s_1), where the 367 same ambiguity parameter $t \in [0, 1]$ couples reward and dynamics. Both actions at s_1 self-loop with 368 zero reward. Starting from s_0 , let a stationary policy choose a_1 with probability p . Its worst-case 369 discounted value is

$$W(p) = \min_{t \in [0,1]} \frac{p(1+t)}{1 - \beta p q(t)} = \min_{t \in [0,1]} \frac{p(1+t)}{1 - \beta p(1-t)}.$$

370 At discount factor $\beta = 0.9$, evaluating $W(p)$ at three points gives:

$$\begin{aligned} W(0) &= 0, \\ W(5/9) &= \frac{10}{9}, \quad \text{constant in } t, \\ W(5/18) &= \min_{t \in [0,1]} \frac{10(1+t)}{27 + 9t} = \frac{10}{27} \quad (\text{attained at } t = 0). \end{aligned}$$

371 Since $5/18 = \frac{1}{2}(0 + 5/9)$, concavity of W would require:

$$W(5/18) \geq \frac{1}{2}[W(0) + W(5/9)] = \frac{5}{9}.$$

372 But $10/27 < 5/9$, producing a strict rational-valued Jensen violation.

373 Hence W is not concave, and the analogous worst-case value function for this jointly-ambiguous 374 (θ, P) instance is not concave either, so the outer maximization is not a convex program in this regime.

375 ■

376 D.3 Proof of Proposition 1

377 **Seminorm properties.** Nonnegativity is immediate ($\max \geq \min$). Positive homogeneity: $V_v(\mu)$ is 378 linear in v , so for $c > 0$, $\max_\mu V_{cv}(\mu) - \min_\mu V_{cv}(\mu) = c(\max_\mu V_v(\mu) - \min_\mu V_v(\mu)) = c\|v\|_{\mathcal{D}}$. 379 Subadditivity: since sup of a sum is at most the sum of sups and inf of a sum is at least the sum of infs, 380 $\|u + v\|_{\mathcal{D}} = \max_\mu V_{u+v}(\mu) - \min_\mu V_{u+v}(\mu) \leq (\max_\mu V_u(\mu) + \max_\mu V_v(\mu)) - (\min_\mu V_u(\mu) + 381 \min_\mu V_v(\mu)) = \|u\|_{\mathcal{D}} + \|v\|_{\mathcal{D}}$.

382 **Zero set.** $\|v\|_{\mathcal{D}} = 0$ iff $V_v(\mu)$ is constant over $M(P)$ iff $v^\top \Phi^\top (\mu - \nu) = 0$ for every $\mu, \nu \in$ 383 $M(P)$, which is exactly the definition of $v \in \mathcal{A}(P)$. Every shaping direction $c \in S_{\text{pot}}$ satisfies 384 $c^\top \Phi^\top (\mu - \nu) = 0$ for all $\mu, \nu \in M(P)$ by Lemma 1 (the constant $-(1 - \beta)\rho_0^\top \Phi_{\text{pot}}$ cancels in the 385 difference), so $S_{\text{pot}} \subseteq \mathcal{A}(P)$. The containment can be strict: if a state s is unreachable under every 386 policy then $\mu(s, \cdot) = 0$ throughout $M(P)$, so any v whose induced reward is supported on s lies

in $\mathcal{A}(P)$, while such a reward need not be a potential-shaping term. We do not claim strictness in general, only that $\mathcal{A}(P)$ is the exact zero set of $\|\cdot\|_{\mathcal{D}}$, which is what the invariance claims use.

Regret bound. With μ_{θ}^* optimal for θ and $\mu_{\theta'}$ optimal for θ' ,

$$\begin{aligned} \text{Reg}_{\theta}(\pi_{\theta'}) &= V_{\theta}(\mu_{\theta}^*) - V_{\theta}(\mu_{\theta'}) \\ &= [V_{\theta}(\mu_{\theta}^*) - V_{\theta'}(\mu_{\theta}^*)] - [V_{\theta}(\mu_{\theta'}) - V_{\theta'}(\mu_{\theta'})] + [V_{\theta'}(\mu_{\theta}^*) - V_{\theta'}(\mu_{\theta'})] \\ &= V_{\theta-\theta'}(\mu_{\theta}^*) - V_{\theta-\theta'}(\mu_{\theta'}) + \underbrace{[V_{\theta'}(\mu_{\theta}^*) - V_{\theta'}(\mu_{\theta'})]}_{\leq 0 \text{ since } \mu_{\theta'} \text{ maximizes } V_{\theta'} \text{ over } M(P)} \\ &\leq V_{\theta-\theta'}(\mu_{\theta}^*) - V_{\theta-\theta'}(\mu_{\theta'}) \leq \max_{\mu} V_{\theta-\theta'}(\mu) - \min_{\mu} V_{\theta-\theta'}(\mu) = \|\theta - \theta'\|_{\mathcal{D}}, \end{aligned}$$

the last step because $\mu_{\theta}^*, \mu_{\theta'} \in M(P)$, so their $V_{\theta-\theta'}$ values each lie in $[\min_{\mu} V_{\theta-\theta'}(\mu), \max_{\mu} V_{\theta-\theta'}(\mu)]$.

Reparametrization invariance. For invertible A , writing $\tilde{\phi} = A\phi$, $\tilde{v} = A^{-\top}v$: $\tilde{\phi}(s, a)^{\top} \tilde{v} = \phi(s, a)^{\top} A^{\top} A^{-\top} v = \phi(s, a)^{\top} v$ for every (s, a) , so the reward function is unchanged, hence $V_v(\mu)$ at every $\mu \in M(P)$, hence $\|v\|_{\mathcal{D}}$ itself. ■

D.4 Proof of Proposition 4

On the quotient by $\mathcal{A}(P)$, $\|\cdot\|_{\mathcal{D}}$ is a norm and every nonnull positive ray has a unit representative. Hence the triangle inequality, symmetry, and nonnegativity of d_o follow from the seminorm, and $d_o \leq \|\theta^o\|_{\mathcal{D}} + \|\theta'^o\|_{\mathcal{D}} = 2$. Independent positive rescaling does not change either unit representative. Adding an element of $\mathcal{A}(P)$ changes a representative only by a seminorm-zero vector, and feature invariance follows from Proposition 1.

If $d_o(\theta, \theta') = 0$, then $\theta/\|\theta\|_{\mathcal{D}} - \theta'/\|\theta'\|_{\mathcal{D}} \in \mathcal{A}(P)$; rearranging gives $\theta' = c\theta + a$ with $c = \|\theta'\|_{\mathcal{D}}/\|\theta\|_{\mathcal{D}} > 0$ and $a \in \mathcal{A}(P)$. The converse is immediate. Finally apply Proposition 1 to the two normalized rewards. Their optimal policies are unchanged by positive normalization, so

$$\text{Reg}_{\theta}(\pi_{\theta'})/\|\theta\|_{\mathcal{D}} = \text{Reg}_{\theta^o}(\pi_{\theta'^o}) \leq \|\theta^o - \theta'^o\|_{\mathcal{D}} = d_o(\theta, \theta'). \quad \blacksquare$$

D.5 Proof of Theorem 1

Equivalence. By definition $s_k = \min_{\theta \in K(\pi_k)} \|\theta - \bar{\theta}^o\|_{\mathcal{D}}$, so $s_k \leq q$ iff some $\theta \in K(\pi_k)$ achieves $\|\theta - \bar{\theta}^o\|_{\mathcal{D}} \leq q$, i.e. iff $K(\pi_k) \cap B_q \neq \emptyset$. This is an immediate consequence of s_k 's definition as a minimum over the cone, not a separate argument. The epigraph formulation is a feasible, bounded-below LP over a closed polyhedron, so the fundamental theorem of linear programming gives an attained optimum. The implementation adds a large numerical safety cap and rejects a solution if that cap is active.

Coverage. $\bar{\theta}$ is a fixed, measurable function of the training split, disjoint from the calibration split and the fresh demonstrator. Conditional on $\bar{\theta}$, each s_k is a deterministic, measurable function of one i.i.d. demonstrator draw, so $s_1, \dots, s_{N_{\text{val}}}, s_{\text{new}}$ are exchangeable. The ascending rank of s_{new} among these $N_{\text{val}} + 1$ exchangeable reals is therefore uniform on $\{1, \dots, N_{\text{val}} + 1\}$, so $\mathbb{P}[\text{rank}(s_{\text{new}}) \leq \tau] = \tau/(N_{\text{val}} + 1) \geq \gamma$ by the choice of τ ; $s_{\text{new}} \leq q_{\gamma}$ iff $\text{rank}(s_{\text{new}}) \leq \tau$ (ties broken uniformly at random, as in the proof of Theorem 4). ■

D.6 Proof of Theorem 2

On the Stage-I event, choose a witness $\tilde{\theta} \in K(\pi_{\text{new}})$ with $\|\tilde{\theta} - \bar{\theta}^o\|_{\mathcal{D}} \leq q_{\gamma}$. If $q_{\gamma} < 1$, the reverse triangle inequality gives $\|\tilde{\theta}\|_{\mathcal{D}} \geq 1 - q_{\gamma} > 0$. Since $K(\pi)$ is a cone, $\bar{\theta}^o = \tilde{\theta}/\|\tilde{\theta}\|_{\mathcal{D}}$ belongs to $F^o(\pi_{\text{new}})$. Moreover,

$$\begin{aligned} \|\bar{\theta}^o - \bar{\theta}^o\|_{\mathcal{D}} &\leq \|\tilde{\theta}^o - \tilde{\theta}\|_{\mathcal{D}} + \|\tilde{\theta} - \bar{\theta}^o\|_{\mathcal{D}} \\ &= |1 - \|\tilde{\theta}\|_{\mathcal{D}}| + q_{\gamma} \leq 2q_{\gamma}. \end{aligned}$$

The latent normalized reward is in the same fiber, so Assumption 2 and the triangle inequality give $\|\hat{\theta}_{\text{new}}^o - \bar{\theta}^o\|_{\mathcal{D}} \leq \eta + 2q_{\gamma}$. If $q_{\gamma} \geq 1$, the universal projective diameter 2 gives the stated fallback. Theorem 1 assigns probability at least γ to the Stage-I event, proving containment.

424 For the robust statement, μ_{ref} is feasible and has relative value zero under every reward, so the
 425 optimum at μ_R is nonnegative. On the containment event, $\hat{\theta}_{\text{new}}^o \in B_R$; hence its relative value at μ_R
 426 is at least the worst-case relative value and therefore at least zero. Positive rescaling back to $\hat{\theta}_{\text{new}}$
 427 preserves the inequality. Subtracting both values from the same optimal value gives the equivalent
 428 regret statement. ■

429 D.7 Proof of Proposition 2

430 Let $x = \Phi^\top(\mu - \mu_{\text{ref}})/(1 - \beta)$. Polar duality and $\|\bar{\theta}^o\|_{\mathcal{D}} = 1$ give $V_{\bar{\theta}^o}(\mu) - V_{\bar{\theta}^o}(\mu_{\text{ref}}) \leq \gamma_{D_F}(x)$.
 431 Therefore the relative robust objective is at most $(1 - R)\gamma_{D_F}(x) \leq 0$ when $R \geq 1$. The reference has
 432 $x = 0$ and attains zero. If $R > 1$, equality requires $\gamma_{D_F}(x) = 0$, which means the two occupancies
 433 have the same feature vector modulo the null directions; for $R = 1$, additional exposed-face ties can
 434 attain zero. ■

435 D.8 Proof of Theorem 4

436 Under Assumption 1, the sequence of pairs

$$(\hat{\theta}_1, \pi_1), \dots, (\hat{\theta}_{N_{\text{val}}}, \pi_{N_{\text{val}}}), (\hat{\theta}_{\text{new}}, \pi_{\text{new}})$$

437 are i.i.d., and therefore exchangeable. The calibration score

$$c_k = \max_{\theta \in \hat{\Theta}_2(\pi_k)} \theta^\top \bar{\theta}$$

438 (Proposition 3) is a fixed measurable function of $(\hat{\theta}_k, \pi_k)$ once $\bar{\theta}$ is fixed (computed on a disjoint
 439 training split, independent of the calibration and test data). Consequently, the calibration scores

$$c_1, \dots, c_{N_{\text{val}}}, c_{\text{new}}$$

440 are themselves exchangeable.

441 Break ties at random only for the rank argument (the implemented inclusive inequality remains conser-
 442 vative under ties). Exchangeability makes the descending rank of c_{new} uniform on $\{1, \dots, N_{\text{val}} + 1\}$.
 443 If $\tau \leq N_{\text{val}}$, this gives

$$\mathbb{P}(c_{\text{new}} \geq q_\gamma) \geq \frac{\tau}{N_{\text{val}} + 1} \geq \gamma.$$

444 If $\tau = N_{\text{val}} + 1$, then $q_\gamma = -1$ and the event holds surely because $\|\bar{\theta}\|_2 = 1$ and every score lies in
 445 $[0, 1]$ (the zero reward is feasible for every policy).

446 It remains to show that the event $\{c_{\text{new}} \geq q_\gamma\} = \{c_{\text{new}} \geq \cos \alpha_\gamma\}$ is equivalent to $\hat{\Theta}_{2,\text{new}} \cap$
 447 $C(\bar{\theta}, \alpha_\gamma) \neq \emptyset$:

$$\begin{aligned} c_{\text{new}} \geq \cos \alpha_\gamma &\iff \exists \theta \in \hat{\Theta}_{2,\text{new}} \text{ such that } \theta^\top \bar{\theta} \geq \cos \alpha_\gamma \\ &\iff \exists \theta \in \hat{\Theta}_{2,\text{new}} \text{ such that } \theta \in C(\bar{\theta}, \alpha_\gamma) \\ &\iff \hat{\Theta}_{2,\text{new}} \cap C(\bar{\theta}, \alpha_\gamma) \neq \emptyset. \end{aligned}$$

448 Combining these statements yields:

$$\mathbb{P}(\hat{\Theta}_{2,\text{new}} \cap C(\bar{\theta}, \alpha_\gamma) \neq \emptyset) \geq \gamma. \quad \blacksquare$$

449 D.9 Proof of Theorem 5

450 Write

$$V_\theta(\mu) := \frac{\theta^\top \Phi^\top \mu}{1 - \beta}$$

451 for the expected discounted return of occupancy measure μ under reward θ . By Cauchy–Schwarz,
 452 this function is $\bar{\nu}$ -Lipschitz in θ (in ℓ_2) uniformly over $\mu \in M(P)$:

$$\begin{aligned} |V_\theta(\mu) - V_{\theta'}(\mu)| &= \frac{|(\theta - \theta')^\top \Phi^\top \mu|}{1 - \beta} \\ &\leq \|\theta - \theta'\|_2 \frac{\|\Phi^\top \mu\|_2}{1 - \beta} \\ &\leq \|\theta - \theta'\|_2 \nu(\mu). \end{aligned}$$

Let $\theta_*^- \in \arg \min_{\theta \in C} V_\theta(\mu^*)$ and $\theta_{\text{CIRL}}^- \in \arg \min_{\theta \in C} V_\theta(\mu_{\text{CIRL}})$; compactness of C ensures both exist. Robust optimality gives the correctly quantified comparison

$$V_{\theta_{\text{CIRL}}^-}(\mu_{\text{CIRL}}) \geq V_{\theta_*^-}(\mu^*).$$

For $\alpha \leq \pi/2$, the Euclidean diameter of the unit-ball cap is at most $2 \sin \alpha$: its extreme separation occurs between two unit vectors on the rim whose angular separation is 2α . Because θ_1, θ_*^- , and θ_{CIRL}^- all lie in C , Assumption (ii) and the triangle inequality imply

$$\|\theta_*^- - \theta_{\text{new}}^*\|_2, \|\theta_{\text{CIRL}}^- - \theta_{\text{new}}^*\|_2 \leq 2 \sin \alpha + \eta.$$

We may therefore decompose

$$\begin{aligned} \text{AOG} &= V_{\theta_{\text{new}}^*}(\mu^*) - V_{\theta_{\text{new}}^*}(\mu_{\text{CIRL}}) \\ &\leq [V_{\theta_{\text{new}}^*}(\mu^*) - V_{\theta_*^-}(\mu^*)] + [V_{\theta_{\text{CIRL}}^-}(\mu_{\text{CIRL}}) - V_{\theta_{\text{new}}^*}(\mu_{\text{CIRL}})] \\ &\leq (\eta + 2 \sin \alpha)(\nu(\mu^*) + \nu(\mu_{\text{CIRL}})). \end{aligned}$$

The uniform corollary follows by replacing both local Lipschitz constants with $\bar{\nu}$. ■

D.10 Proof of Proposition 5

If $\alpha = 0$, the cap is the singleton $\{\bar{\theta}\}$ and both claims are immediate. Assume henceforth that $0 < \alpha < \pi/2$.

Fix $x \in \mathbb{R}^d$ and write $a = \bar{\theta}^\top x$, $x_\perp = x - a\bar{\theta}$, $b = \|x_\perp\|_2$, so $\|x\|_2^2 = a^2 + b^2$. Any θ with $\|\theta\|_2 \leq 1$ decomposes as $\theta = u\bar{\theta} + v\hat{w}$ with $\hat{w} \perp \bar{\theta}$ a unit vector, $u = \bar{\theta}^\top \theta$, $v \geq 0$, and $u^2 + v^2 = \|\theta\|_2^2 \leq 1$. The cap constraint $\theta^\top \bar{\theta} \geq \cos \alpha$ reads $u \geq \cos \alpha$. Since $\theta^\top x = ua + v\hat{w}^\top x_\perp$ and $\hat{w}$ enters only through $\hat{w}^\top x_\perp \in [-b, b]$, the minimizing choice is $\hat{w} = -x_\perp/b$ when $b > 0$ (any $\hat{w}$ when $b = 0$), giving $\theta^\top x = au - bv$. The inner problem is therefore the two-variable program

$$\min \{ au - bv : u \geq \cos \alpha, v \geq 0, u^2 + v^2 \leq 1 \}.$$

As $b \geq 0$, it is optimal to take v maximal for the chosen u , i.e. $v = \sqrt{1 - u^2}$, leaving $\min_{u \in [\cos \alpha, 1]} f(u)$ with $f(u) := au - b\sqrt{1 - u^2}$ and $f'(u) = a + bu/\sqrt{1 - u^2}$, which is strictly increasing in u on $(-1, 1)$. Hence f is either increasing on the whole interval (the constraint $u \geq \cos \alpha$ binds) or has a unique interior stationary point.

Case 1: $f'(\cos \alpha) \geq 0$, i.e. $a + b \cot \alpha \geq 0$. Then f is nondecreasing on $[\cos \alpha, 1]$ and the minimum is at $u = \cos \alpha$, $v = \sin \alpha$, giving $f = a \cos \alpha - b \sin \alpha$.

Case 2: $f'(\cos \alpha) < 0$. Setting $f'(u^*) = 0$ gives $u^*/\sqrt{1 - u^{*2}} = -a/b =: c > 0$, so $u^* = c/\sqrt{1 + c^2}$ and $f(u^*) = (ac - b)/\sqrt{1 + c^2} = -\sqrt{a^2 + b^2} = -\|x\|_2$, the unconstrained minimum over the unit ball, attained at $\theta = -x/\|x\|_2$. This is feasible exactly when $-x/\|x\|_2$ lies in the cap, i.e. $a \leq -\|x\|_2 \cos \alpha$, which is equivalent to $f'(\cos \alpha) < 0$ after squaring. The two cases agree at the boundary $a = -\|x\|_2 \cos \alpha$, where $b = \|x\|_2 \sin \alpha$ and $a \cos \alpha - b \sin \alpha = -\|x\|_2(\cos^2 \alpha + \sin^2 \alpha) = -\|x\|_2$, so the stated piecewise formula is continuous.

For the second claim, every nonzero $x = \Phi^\top \mu$ with $a(x) \geq 0$ lies in Case 1, while at $x = 0$ both displayed branches equal zero. Hence the robust objective is $\mu \mapsto a \cos \alpha - b \sin \alpha$ uniformly on $M(P)$. Since $\alpha < \pi/2$ gives $\cos \alpha > 0$, dividing the objective by the positive constant $\cos \alpha$ leaves the arg max unchanged and yields $a - \tan \alpha b$, which is the stated program. Concavity is immediate from this form: $a(\Phi^\top \mu) = \bar{\theta}^\top \Phi^\top \mu$ is linear in μ , while $b(\Phi^\top \mu)$ is a seminorm composed with a linear map, hence convex, and $\tan \alpha \geq 0$. ■

D.11 Proof of Lemma 1

Expand, using the definition of a shaping direction,

$$\begin{aligned} c^\top \Phi^\top \mu &= \sum_{s,a} \mu(s,a) [\beta P(s,a,\cdot)^\top \Phi_{\text{pot}} - \Phi_{\text{pot}}(s)] \\ &= \sum_{s'} \Phi_{\text{pot}}(s') \left[\beta \sum_{s,a} P(s,a,s') \mu(s,a) \right] - \sum_s \Phi_{\text{pot}}(s) \sum_a \mu(s,a). \end{aligned}$$

488 The flow-conservation constraint defining $M(P)$ gives, for every s' , $\sum_a \mu(s', a) -$
 489 $\beta \sum_{s,a} P(s, a, s') \mu(s, a) = (1 - \beta) \rho_0(s')$, i.e. $\beta \sum_{s,a} P(s, a, s') \mu(s, a) = \sum_a \mu(s', a) - (1 -$
 490 $\beta) \rho_0(s')$. Substituting into the first term,

$$c^\top \Phi^\top \mu = \underbrace{\sum_{s'} \Phi_{\text{pot}}(s') \sum_a \mu(s', a)}_{=\sum_s \Phi_{\text{pot}}(s) \sum_a \mu(s, a)} - (1 - \beta) \sum_{s'} \Phi_{\text{pot}}(s') \rho_0(s') - \sum_s \Phi_{\text{pot}}(s) \sum_a \mu(s, a),$$

491 and the first and third terms are identical, hence cancel, leaving $c^\top \Phi^\top \mu = -(1 - \beta) \rho_0^\top \Phi_{\text{pot}}$,
 492 independent of μ . This uses only the linear flow-conservation constraints defining $M(P)$, so it holds
 493 at every $\mu \in M(P)$, not only at policy-occupancy vertices. ■

494 D.12 Proof of Theorem 6

495 By Lemma 1, $\theta'^\top \Phi^\top \mu = \bar{\theta}^\top \Phi^\top \mu + c^\top \Phi^\top \mu = \bar{\theta}^\top \Phi^\top \mu - (1 - \beta) \rho_0^\top \Phi_{\text{pot}}$ for every $\mu \in M(P)$,
 496 so the two objectives $\mu \mapsto \bar{\theta}^\top \Phi^\top \mu$ and $\mu \mapsto \bar{\theta}'^\top \Phi^\top \mu$ differ by a μ -independent additive constant.
 497 Hence they share the same maximizer, $\mu_{\bar{\theta}'}^* = \mu_{\bar{\theta}}^*$, and $V_{\bar{\theta}'}(\mu_{\bar{\theta}'}^*) = V_{\bar{\theta}}(\mu_{\bar{\theta}}^*) - \rho_0^\top \Phi_{\text{pot}}$. The same
 498 additive-constant identity applied at $\mu = \mu_{\pi_k}$ gives $V_{\bar{\theta}'}(\mu_{\pi_k}) = V_{\bar{\theta}}(\mu_{\pi_k}) - \rho_0^\top \Phi_{\text{pot}}$. Subtracting,

$$c_k(\bar{\theta}') = [V_{\bar{\theta}}(\mu_{\bar{\theta}}^*) - \rho_0^\top \Phi_{\text{pot}}] - [V_{\bar{\theta}}(\mu_{\pi_k}) - \rho_0^\top \Phi_{\text{pot}}] = c_k(\bar{\theta}). \quad \blacksquare$$

499 D.13 Proof of Proposition 6

500 Write $\theta = \bar{\theta} + \delta$, $\|\delta\|_2 \leq \rho$. Then $\theta^\top x = \bar{\theta}^\top x + \delta^\top x$, and $\min_{\|\delta\|_2 \leq \rho} \delta^\top x = -\rho \|x\|_2$ by
 501 Cauchy–Schwarz (equality at $\delta = -\rho x / \|x\|_2$ when $x \neq 0$, and trivially at $x = 0$), giving
 502 $\min_{\theta: \|\theta - \bar{\theta}\|_2 \leq \rho} \theta^\top x = \bar{\theta}^\top x - \rho \|x\|_2$ for every x , with no case split. Substituting $x = \Phi^\top \mu$ gives
 503 the stated robust MDP objective directly; both terms are defined for every $\bar{\theta}, \rho \geq 0, \mu \in M(P)$, so
 504 no sign condition on $\bar{\theta}^\top \Phi^\top \mu$ is needed (contrast Proposition 5, whose regularizer form requires the
 505 nonnegative-return branch). ■